

Symbolic Audit of the Implemented χ/W Surface for the $H \parallel c$, $E \parallel a$ Readout in TmFeO₃

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1 Statement of the audit

The question is whether the *currently implemented and currently scanned* mixed Fe–Tm Hamiltonian can generate a physically meaningful Fe–Tm crosspeak at excitation near the Fe qAFM mode and detection near the Tm $E_1 \rightarrow E_2$ line, when the experimental readout is taken from the stored mixed-lattice observable `M_global_SU3` in its `lambda1/lambda2` components.

The working criterion is the same as in the previous note: amplitudes of order 10^{-3} or smaller are not counted as genuine.

The conclusion remains negative, but the correct reason is slightly sharper than the markdown draft: the obstruction is not that one is measuring a naive uniform sum of raw local λ_1 and λ_2 . The real obstruction is that the code measures an *induced-frame* global SU(3) observable, and in the present Geometry-II setting the implemented χ source, the CEF Liouvillian, and the scanned W_1^{xy}/W_1^{yz} vertices never generate the four-sublattice character required by that readout.

2 The actual code-level readout operator

In the mixed Tm sector the active magnetic and electric SU(3) triplets are

$$\mathbf{m}_\mu = \begin{pmatrix} \lambda_\mu^2 \\ \lambda_\mu^5 \\ \lambda_\mu^7 \end{pmatrix}, \quad \mathbf{e}_\mu = \begin{pmatrix} \lambda_\mu^1 \\ \lambda_\mu^4 \\ \lambda_\mu^6 \end{pmatrix}, \quad (1)$$

with one triplet per Tm sublattice $\mu = 1, \dots, 4$.

The code does *not* store a raw local-site sum. Instead it uses the induced frame

$$\begin{aligned} \mathcal{F}_\mu &= \mu_{\text{act}}^{-1} R_\mu \mu_{\text{act}}, \\ R_1 &= \text{diag}(+, +, +), \quad R_2 = \text{diag}(+, -, -), \\ R_3 &= \text{diag}(-, +, -), \quad R_4 = \text{diag}(-, -, +), \end{aligned} \quad (2)$$

with the same 3×3 active block acting on both $\mathbf{m}_\mu$ and $\mathbf{e}_\mu$.

The stored global observables are therefore

$$\mathbf{M}_{\text{global}}^{(m)} = \frac{1}{4} \sum_{\mu=1}^4 \mathcal{F}_\mu \mathbf{m}_\mu, \quad \mathbf{M}_{\text{global}}^{(e)} = \frac{1}{4} \sum_{\mu=1}^4 \mathcal{F}_\mu \mathbf{e}_\mu, \quad (3)$$

and the saved components labelled `lambda2` and `lambda1` are just

$$M_{\text{global}}[\lambda_2] = \mathbf{u}_1^\top \mathbf{M}_{\text{global}}^{(m)}, \quad M_{\text{global}}[\lambda_1] = \mathbf{u}_1^\top \mathbf{M}_{\text{global}}^{(e)}, \quad (4)$$

with $\mathbf{u}_1 = (1, 0, 0)^\top$.

This matters because Eqs. (3)–(4) already include the four-sublattice sign structure. For example, the default projected moment matrix makes the c -axis magnetic probe proportional to the stored global λ_2 component, which in local language is the $(+, -, -, +)$ -weighted sum of raw local λ_2 values, not a naive uniform sum.

3 Implemented Hamiltonian in the present scan surface

At the level relevant here, the mixed model is

$$H = H_{\text{CEF}} + H_\chi + H_W + H_{\text{drive}}, \quad (5)$$

with

$$H_{\text{CEF}} = \sum_{\mu=1}^4 (\alpha \lambda_\mu^3 + \beta \lambda_\mu^8), \quad (6)$$

$$H_\chi = \sum_{i,\mu} S_i^a \chi_{ab}(i, \mu) \lambda_\mu^b, \quad b \in \{2, 5, 7\}, \quad (7)$$

$$H_W = \sum_{i,\mu} \lambda_\mu^{(n)} W_n^{ab}(i, \mu) S_i^a S_i^b, \quad n \in \{1, 3, 4, 6, 8\}. \quad (8)$$

The code-level support is therefore:

- χ populates only the magnetic triplet $(\lambda_2, \lambda_5, \lambda_7)$,
- W never populates λ_2 at all,
- the present scan surface used in the existing 2DCS campaign turns on pure χ lines and the two even W_1 lines W_1^{xy} and W_1^{yz} .

The odd $W_{4,6}$ channels exist in the builder, but they are not part of the current scan surface audited here.

4 Klein-four character bookkeeping and the SU(3) EOM

For the symbolic audit it is convenient to expand any four-sublattice pattern in the Klein-four character basis

$$\sigma_F = (+, +, +, +), \quad \sigma_C = (+, +, -, -), \quad \sigma_A = (+, -, +, -), \quad \sigma_G = (+, -, -, +). \quad (9)$$

For any site-resolved SU(3) component x_μ^a we write

$$x_\mu^a = \sum_{X \in \{F, C, A, G\}} \sigma_X(\mu) x_X^a, \quad x_X^a = \frac{1}{4} \sum_{\mu=1}^4 \sigma_X(\mu) x_\mu^a. \quad (10)$$

This is only a bookkeeping decomposition of the four-site pattern. It is not a claim that the full Tm sector should always be reduced to a single Fe-like Bertaut variable.

With $[\lambda^a, \lambda^b] = 2if^{abc}\lambda^c$, the local SU(3) equation of motion takes the form

$$\dot{\ell}_\mu^a = \kappa f^{abc} \ell_\mu^b h_\mu^c, \quad (11)$$

where κ is the overall convention-dependent prefactor and $h_\mu^c = -\partial H / \partial \ell_\mu^c$.

Substituting Eq. (10) into Eq. (11) gives the character-resolved equation

$$\dot{\ell}_Z^a = \kappa \sum_{\substack{X,Y \in \{F,C,A,G\} \\ XY=Z}} f^{abc} \ell_X^b h_Y^c, \quad (12)$$

where the multiplication is the Klein-four group law coming from pointwise sign products, for example $G \cdot A = C$ and $G \cdot G = F$.

Equation (12) is the key symbolic audit. It implies three facts immediately:

1. An F -character field preserves the four-sublattice character, because $F \cdot X = X$.
2. To convert a G -character coherence into a C -character coherence one needs an A -character field, because $G \cdot A = C$.
3. The subset $\{F, G\}$ is closed under multiplication, so a dynamics built only from F - and G -character fields can never generate a C -character response.

The CEF field in Eq. (6) is uniform, hence pure F -character. Therefore the linearized CEF Liouvillian is block diagonal in the character label. In particular, on the $1 \leftrightarrow 2$ coherence pair one has schematically

$$\partial_t \begin{pmatrix} \ell_X^1 \\ \ell_X^2 \end{pmatrix} = \begin{pmatrix} 0 & +\Omega_{12} \\ -\Omega_{12} & 0 \end{pmatrix} \begin{pmatrix} \ell_X^1 \\ \ell_X^2 \end{pmatrix} + \dots, \quad (13)$$

with the *same* character label X on both components. The CEF can rotate λ_2 into λ_1 , but it cannot change the four-sublattice character while doing so.

5 Geometry-II source channels

For the readout geometry of interest, the derivation already recorded in `docs/tmfeo3_notes.tex` gives

$$H_{\text{drive}}^{m,\text{Tm}} \propto h_z \Lambda_G^2, \quad H_{\text{drive}}^{e,\text{Tm}} \propto e_x \Lambda_C^1. \quad (14)$$

Thus the desired Fe $\rightarrow$ Tm crosspeak requires the following symbolic sequence:

$$\begin{aligned} &\text{qAFM magnetic leg: } G\text{-character in the magnetic triplet} \\ &\longrightarrow \text{electric readout leg: } C\text{-character in the electric triplet.} \end{aligned} \quad (15)$$

From Eq. (12), this means that some implemented vertex must provide an *A-character* effective field while also supplying the triplet conversion needed to turn the magnetic λ_2 -type coherence into the electric λ_1 -type readout direction.

6 Why the present chi plus scanned W1 surface cannot do this

6.1 Chi alone

The magnetic χ tensor populates only the magnetic triplet and injects the qAFM-coupled λ_2 channel. Equation (13) allows that source to rotate locally into a λ_1 -type coherence, but only inside the same character block. Therefore a G -character χ source can only generate G -character λ_1/λ_2 response. It cannot generate the required C -character electric readout of Eq. (14).

So χ provides the right frequency pair, but the wrong character sector.

6.2 W1xy and W1yz

The present scan surface turns on only the even W_1 channel. Linearizing its Fe bilinear around the Γ_2 background

$$\Gamma_2 : \quad F_x \oplus G_z, \quad (16)$$

and the qAFM fluctuation

$$\delta Q_{\text{qAFM}} \sim F_z \oplus G_y, \quad (17)$$

gives the character content

$$S_x \delta S_y : F \cdot G = G, \quad S_z \delta S_y : G \cdot G = F. \quad (18)$$

Hence, at the level linear in the qAFM normal coordinate,

- W_1^{xy} furnishes only a G -character electric-triplet source,
- W_1^{yz} furnishes only an F -character electric-triplet source.

Neither one supplies the missing A -character vertex. Since the CEF field is also F -character, the entire symbolic closure of the present $\chi + W_1^{xy} + W_1^{yz} + H_{\text{CEF}}$ surface is

$$\{F, G\} \cdot \{F, G\} = \{F, G\}, \quad (19)$$

which never reaches the C block required by the $E \parallel a$ readout.

This is the cleanest symmetry statement of the obstruction: with the current scan surface, the EOM never leaves the $\{F, G\}$ subgroup, while the target readout needs the C block.

6.3 Why this is enough to rule out the observed crosspeak

The desired third-order signal is not merely a Tm resonance near ω_{12} . It is specifically a Geometry-II Fe→Tm crosspeak: qAFM excitation on the magnetic leg and electric-triplet readout on the Tm leg.

The current implementation fails to provide the required symbolic bridge:

1. χ provides the qAFM-active magnetic source but remains trapped in the wrong character block.
2. W_1 provides electric-triplet weight but only in F or G , never in the C block demanded by $E \parallel a$.
3. The uniform CEF field preserves character and therefore cannot repair the mismatch.

Therefore the present Hamiltonian does not contain a path from the qAFM-fed Tm sector into the actual induced-frame Geometry-II readout operator.

7 Consistency with the existing 2DCS data

The symbolic audit matches the cached spectra already generated in the repo. For the baselines and for the full W_1^{xy}/W_1^{yz} grid, `M_NL_SU3_FF_lambda1` and `M_NL_SU3_FF_lambda2` are exactly zero at the nearest grid points to

$$\omega_\tau = 0.911105 \text{ THz}, \quad \omega_t = 0.493319 \text{ THz}, \quad \omega_t = 0.693729 \text{ THz}. \quad (20)$$

The separate raw-state control also remains consistent with the EOM audit: it can hide tiny weight in a non-measured character channel, but that weight is only 6.99×10^{-5} at (0.911, 0.493) THz and therefore still far below the 10^{-3} threshold.

8 Final verdict

For the current TmFeO₃ implementation and the current scan surface, the absence of the Fe–Tm crosspeak is structural.

The decisive symbolic facts are:

1. the code-level readout lives in the induced-frame observable `M_global_SU3`, not in a raw local-site sum,
2. the Geometry-II source channels require a $G \rightarrow C$ transfer in the Tm sector,
3. the present χ plus scanned W_1 surface supplies only F - and G -character vertices, while the CEF Liouvillian preserves character,
4. therefore the linearized SU(3) EOM never reaches the required C block.

So the observed qAFM $\rightarrow E_1 \rightarrow E_2$ Fe–Tm crosspeak cannot be produced by retuning the existing χ coefficients or the current W_1^{xy}/W_1^{yz} scan line alone. To obtain that signal, the model needs an additional interaction or readout channel that injects the missing character structure into the electric triplet in the induced-frame basis.